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Hydrogen superpermeation resistant to ion sputtering

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A nonmetallic monolayer on the surface of a metallic membrane gives rise to the superpermeation of suprathermal hydrogen. This means that virtually the whole implanted flux passes through the membrane, not depending on membrane temperature and thickness. The fact that superpermeation is degrading at a sufficiently high energy of incident hydrogen particles was considered a ''natural'' result of sputtering of the monolayer. It is now proven that Nb containing an O impurity would be normally superpermeable to suprathermal hydrogen *at any incident energy*, whereas the ''usual'' sensitivity to sputtering is conditioned by a NbC layer originally present at the ''real'' surface of Nb. © *2002 American Institute of Physics.* [DOI: 10.1063/1.1510961]

A ''real'' metallic surface in vacuum is typically covered with a nonmetallic monolayer. For example, if a metal is subjected to heating in vacuum, virgin polyatomic films are desorbed or dissolved, only a monolayer of nonmetal impurity remaining at the surface. The monolayer gives rise to a potential barrier impeding both the dissociative absorption and associative desorption of *thermal* H_2 molecules.[1] However, if the metal is acted upon by *suprathermal* hydrogen particles whose energy (kinetic, internal, or chemical) exceeds ~1 eV, the barrier does not prevent their absorption, while continuing to suppress the recombinative release of H atoms thermalized in the lattice. That increases hydrogen retention in, and permeation through, the metal by many orders of magnitude.[1] The re-emission may be impeded to such an extent that the implanted H atoms cross a plane membrane sample many times before the release. As a result, if the two membrane surfaces are identical (most typical case), *~1/2 of the implanted flux passes through the membrane, irrespective of its temperature and thickness*. Thus the membrane becomes ''superpermeable'', that is, almost ''transparent'' to suprathermal hydrogen.[1]

Given this role of nonmetallic monolayers, one naturally would expect sputtering to produce a significant effect. In practice, a ''permeation spike'' is typically observed: after the initiation of a flux of hydrogen ions of an energy, E_{inc}, at keV level hitting a membrane of V,[2] Ni, Mo[3] or stainless steel,[1] the permeation flux first passes a maximum, but then gradually decreases to eventually reach an orders-of-magnitude lower level.

Energy dependence of the effect of ion bombardment was investigated on Nb[4–6] as one of the best materials (along with V and Ta) for superpermeation. After an elimination of the virgin oxide film in vacuum, oxygen segregates to the niobium surface from metal bulk to form a monolayer coverage (e.g., NbO on Nb[110]).[4–8] At E_{inc} below the oxygen sputtering threshold, the niobium membrane was superpermeable to suprathermal hydrogen, exhibiting all the characteristic features of the phenomenon, including the permeation flux temperature independence.[4–6] With E_{inc} exceeding the oxygen sputtering threshold, oxygen monolayer at the upstream surface gets damaged, which brings about an asymmetry of hydrogen release in favor of re-emission through the upstream surface. As a result, the permeation flux drops quickly (in a few seconds) to a much lower steady state level (Fig. 1: graph 1, and inset). Such a result was obtained at different laboratories on Nb samples of the Russian and Japanese make, both in ion beam[1,9] and plasma (Fig. 1)[4–6] experiments. Since the resistance to sputtering is a key point for applications of superpermeable membranes for particle flux control in nuclear fusion (e.g., Ref. 10), special efforts were made to reduce the effect of sputtering by increasing the rate of surface segregation of oxygen. Both doping the Nb bulk with oxygen and raising the membrane temperature produces a positive effect, but the decrease in permeation flux under sputtering still remained significant.[5,6]

Thus, one was surprised to see the effect of sputtering disappear (Fig. 1, graph 2) in one of our plasma-membrane experiments, with superpermeation remaining stable over an energy range up to at least 1 keV during the whole observation time, that is, in tens of minutes. That happened after an *in-situ* membrane heating at $T=1550$ °C for a few minutes (it

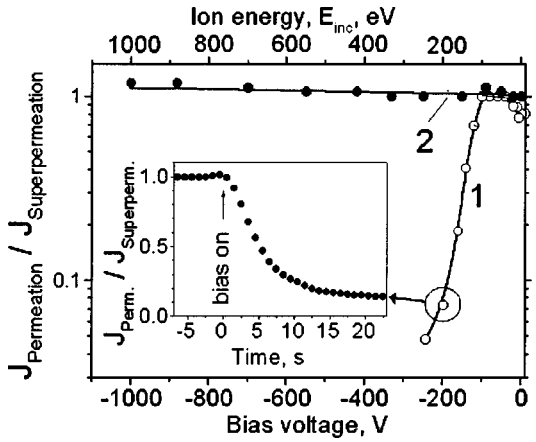


FIG. 1. Dependence of plasma driven permeation on incident ion energy (bias) for two states of the same superpermeable Nb membrane.

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had only been heated to $T \leqslant 1280\,^{\circ}\mathrm{C}$ before). Note that the permeation flux does not depend on E_{inc} in the state resistant to sputtering, even though the implantation coefficient of incident ions does. In fact, it is H atoms of *constant energy* that comprise a major portion of the incident flux of suprathermal hydrogen from plasma, with ions of the varied energy changing permeability to all the types of suprathermal particles (including H atoms) through affecting the nonmetallic monolayer.

A hypothesis to explain the phenomenon was that *responsible for the effect is the decarbonization of the* Nb *sample*. Such a supposition was based on the following: (1) For the first time in our experiments the membrane had been heated above the critical temperature ($\sim 1400\,^{\circ}\mathrm{C}$) of NbC decomposition into Nb and CO in a reaction with solute O.[11] (2) A layer-by-layer Auger electron spectroscopy analysis shows that the original surface of the manufactured Nb foils is typically covered with a polyatomic carbide layer. In contrast to the easy dissolution of virgin oxides, *polyatomic* $\mathrm{NbC/Nb_2C}$ layers were found to remain at both sides of the membrane sample during the whole experimental campaign, even despite a significant heating ($1280\,^{\circ}\mathrm{C}$). A near-monolayer O coverage was found to be present atop the $\mathrm{NbC/Nb_2C}$ layer after the sample had been heated in vacuum at $T \approx 500-600\,^{\circ}\mathrm{C}$. (3) Hydrogen transport even through rather thick (e.g., $\sim 1\ \mu$m) NbC layers has been recently shown[12] to occur quickly enough so as to not impair the Nb superpermeability. (4) With all the above accounted for, one must surmise that either surface of the Nb membrane had a monolayer of O, most probably, upon a polyatomic carbide, prior to membrane heating to $T > 1400\,^{\circ}\mathrm{C}$. The segregation of O onto the carbide (very much like segregation onto a metal surface) may occur due to the metallic properties of NbC.

The hypothesis was examined with the apparatus with which the results of Fig. 1 had been obtained.[4,6] An ohmically heated tubular membrane of polycrystalline Nb (1 cm diameter, 18 cm length, 0.01 cm wall thickness) is immersed in a uniform H plasma (density $\sim 10^{11}\ \mathrm{cm}^{-3}$, electron temperature ~ 1 eV, and $\mathrm{H_2}$ pressure 3×10^{-3} Torr). The incident ion energy was varied by membrane biasing. Two turbomolecular pumps continuously evacuated the down- and upstream sides of the membrane. The *gas impurity* pressure in plasma measured with a differentially pumped quadrupole analyzer was $\sim 1 \times 10^{-6}$ Torr (mainly CO).

The first step was *a controllable decarbonization of the membrane specimen having a natural surface*. First the membrane was heated to a maximum temperature at which no CO release was yet occurring ($T = 1270\,^{\circ}\mathrm{C}$). After that the plasma-driven superpermeation with the *usual* effects of ion energy (similarly to shown in Fig. 1, graph 1) was observed. The membrane temperature was then raised to $1550\,^{\circ}\mathrm{C}$, and a release of CO occurred. The initial CO flux amounted to $\sim 3 \times 10^{15}\ \mathrm{CO/cm^2\,s}$, decreasing $\sim$fivefold in ~ 10 min. The total amount of the released C (and O) was as large as ~ 0.2 at. % (~ 800 monolayers). Resulting from this procedure, the membrane passed into a state resistant to sputtering (Fig. 2), similar to what had happened previously.

The next step was a *direct examination of the effects of the presence of a carbide layer*. To this end, a C layer

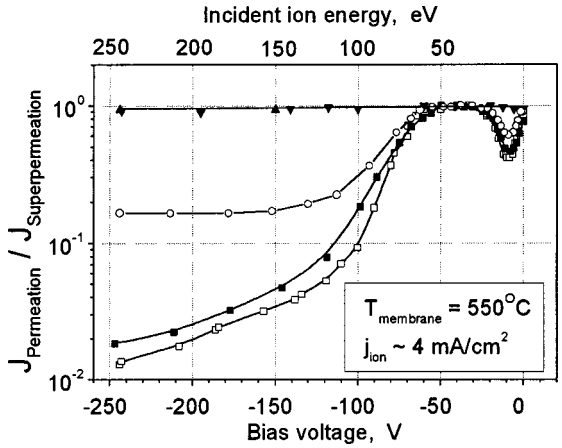


FIG. 2. Dependence of plasma driven permeation on incident ion energy (bias voltage) for the same Nb membrane in different states of carbidization: ▲—after the elimination of natural NbC layer, □—after formation of a NbC layer ($\sim 10^3$ monolayers) at the upstream surface, ■, ○, ▼: after the removal of ~20%, 50%, and 100% of the deposited C, respectively.

($\sim 10^{18}\ \mathrm{C/cm^2}$) was deposited *in situ*, from plasma in a mixture of $\mathrm{H_2} + 10\%\mathrm{CH_4}$, onto the upstream surface of a decarbonized membrane resistant to sputtering; the C layer was then converted into carbide by heating to $1100\,^{\circ}\mathrm{C}$. As before,[13] the deposition of polyatomic C onto a relatively cold Nb ($700\,^{\circ}\mathrm{C}$) resulted in a drastic decrease of plasma-driven permeation, with a complete restoration of superpermeability in the energy range *below sputtering threshold* after the carbidization of this layer at membrane heating. However, the membrane proved to be sensitive to sputtering again (Fig. 2), exhibiting all the typical energy dependence peculiarities:[4–6] the non-monotonic behavior around 10 eV (explained by chemical sputtering of O[5]), and an order-of-magnitude drop at $E_{\mathrm{inc}} > 50$ eV.

A step-by-step C removal from the membrane was then effected by way of C conversion into gaseous CO in a reaction with solute O. The effects of sputtering were decreasing in parallel with the C removal (Fig. 2), the membrane going back to the state resistant to sputtering upon complete C elimination.

This seems to be convincing enough a proof that *the "usual" degradation of superpermeability under ion sputtering* must be due to *the polyatomic carbide layers normally present at a natural* Nb *surface* (probably, at surfaces of other carbide-forming metals as well). The question still remains how superpermeation may be stable under sputtering: it would seem that an O monolayer at the upstream surface must be at least damaged (if not completely removed), and hence the symmetry of hydrogen release is to be impaired in favor of re-emission. Another question is what mechanisms lie behind the effects of carbide.

The following qualitative explanation is suggested. Due to a high solubility of O in Nb (~ 1 at. % at our temperatures[11]), oxygen, unlike carbon, is not concentrated in the near surface region, but is uniformly distributed over the sample bulk. When there is no sputtering, the O coverage, Θ_{OS}, of the two membrane surfaces is in equilibrium with the O dissolved in the lattice: $\Theta_{\mathrm{OS}} = \Theta_{\mathrm{OS}}^{\mathrm{eq}}$. Due to a high heat of surface segregation, ΔH_{segr}, of O in Nb,[8] one has $\Theta_{\mathrm{OS}}^{\mathrm{eq}} \approx 1$ (i.e., a monolayer is present) at all the operational T,

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even though a relative bulk concentration of O, Θ_{OB}, is typically 0.001–0.01. Still, there is a certain *equilibrium* concentration of O vacancies in the O monolayer, $\Theta_{VS}^{eq}=1-\Theta_{OS}^{eq}\ll 1$, at any given T and Θ_{OB}. These vacancies apparently serve as centers of thermal hydrogen desorption.[4,5]

The symmetry of hydrogen release is not violated if the approximate equality of O vacancy concentrations at the two surfaces is not perturbed by the upstream side sputtering. For that to take place:

(1) concentrations of the dissolved O near the up- and downstream surfaces, $\Theta_{OB}(0)$ and $\Theta_{OB}(\ell)$, should remain nearly equal:
$$\Theta_{OB}(0)\approx\Theta_{OB}(\ell). \tag{1}$$
(2) the concentration of O vacancies on the upstream surface subject to sputtering, Θ_{VS}, should be staying close to its equilibrium value, that is,
$$\Theta_{VS}\approx\Theta_{VS}^{eq}. \tag{2}$$

Certainly, if even sputtering does not perturb the membrane symmetry, it is gradually exhausting the stored bulk O. However, it is a long-term process (and besides, the membrane may be doped with O *in situ*, if necessary).

Condition (2) is fulfilled when the sputtering rate is much smaller than the rate of the spontaneous O atom transitions from the surface into the lattice:

$$j_{inc}\gamma\Theta_{OS}\ll 10^{15}\Theta_{OS}\nu\exp-\frac{E_{OD}+\Delta H_{segr}}{kT}, \tag{3}$$

where j_{inc} is an incident ion flux density, γ is an O sputtering coefficient, ν is an attempt frequency, and E_{OD} is the activation energy of O diffusion in Nb. For our condition ($j_{inc}\gamma\approx 1\times 10^{14}$ cm^{-2} s^{-1}, $T>550$ °C), and with $\nu\approx 10^{13}$ s^{-1}, $E_{OD}=115$ kJ/mol[11] and $\Delta H_{segr}=72$ kJ/mol,[8] the inequality (3) is fairly well satisfied, and hence Eq. (2) holds.

One can show that condition (1) also is roughly satisfied for our 0.1-mm-thick sample and for a sputtering rate $\sim 10^{14}$ O/cm^2 s due to the relatively high O mobility in Nb ($D\approx 10^{-9}$ cm^2/s at our lowest T, 550 °C[11]). *Thus, the superpermeation resistant against sputtering appears to be a normal property of* Nb *with a natural or specially introduced bulk* O *impurity*.

The drop of permeation under sputtering occurs very quickly, in ~5 s, in the presence of NbC (see inset of Fig. 1). That means that sputtering only *increases the O vacancy concentration at the upstream surface* ($\Theta_{VS}\gg\Theta_{VS}^{eq}$) at *virtually unchanged concentration of solute* O (such a change would have needed a much longer time; for example, a removal of 10% of the bulk O at $\Theta_{OB}=0.002$ would take ~1000 s). Thus one concludes that the NbC layer *impedes* O *exchange between the surface and lattice*, most probably, because of an orders-of-magnitude slower diffusion of O in NbC than in Nb.[14]

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